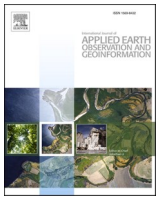
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Assessing the impact of Sentinel-1 derived planting dates on rice crop yield modeling

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ABSTRACT

Planting date (PD) is one of the critical component in crop modeling that has a large-scale impact on seasonal yield estimation. Here, we aimed to estimate the various paddy/rice attributes (particularly, acreage and PD) using the Sentinel-1A observations, and assessing the impact of such key parameters on end-of-season yields. The method comprises: (1) a multi-temporal time-series analysis algorithm of Sentinel-1A observations for rice mapping, (2) a PD retrieval algorithm, and (3) rice yield estimation using the Synthetic Aperture Radar (SAR)-derived PD in a modeling system. Our study was implemented over the rainfed paddy fields in Cambodia between 2017 thru 2020. We demonstrated that the backscatter ratio ($\frac{\sigma_{VH}{}^0}{\sigma_{VV}{}^0}$) from Sentinel-1A can improve the accuracy of rice mapping, while $\sigma_{VH}{}^0$ backscatter can provide PD estimations. Our results showcased reasonable/good performance in mapping the paddy fields and estimating PD in the study area. PD found in March and April were indicative of irrigation application in some parts of Cambodia, specifically in the southeastern part (also part of the Mekong delta) where irrigation in paddy fields is a common practice. Differences between the crop-calendar based PD (June 1) and the SAR-derived PD can range as much as 75 days. In the rice-dominated provinces such as Banteay-Meanchey and Svay-Rieng of Cambodia, when SAR-derived PD are used, rice yield normalized bias improved compared to the fixed PD data by a mean of 7–12 % and 30–48 % respectively within the study period. In addition, the PD derived from SAR also significantly reduced the uncertainty in yield estimation. Overall, this study demonstrated the potential of using the SAR-derived PD in improving the rice yield estimation over rainfed paddy fields.

1. Introduction

Rice is one of the most common sources of staple food for mankind. With 164 million hectares of rice-cultivated land in the year 2020, rice is the third most widely produced cereal in the world (FAO, 2020). Recently rice production is adversely affected globally by extreme weather events (Phung et al., 2020a). Fahad et al. (2018) found that an increase in the frequency/severity of the hot weather could decrease rice production up to 40 % by the end of the 21st century. Therefore, monitoring of paddy over farms is becoming crucial for national food security.

Important geo-bio-physical variables and events such as timely pre-harvest assessment of the PD, crop type detection, acreage estimation, phenology monitoring, and the harvest date determination at a fine spatial resolution are essential for near real-time decision-making at small (farm) to large (country-wide) scale for rice production. Obtaining information on these important variables and events and ingesting them into crop models, can improve the rice yield nowcast and forecast. Most of the rice crop growth models, such as the Decision Support System for Agrotechnology Transfer (DSSAT) (Jones et al., 2003), Agricultural Production Systems Simulator (APSIM) (Keating et al., 2003), CropSyst (Stöckle et al., 2003), Wageningen crop models (van Ittersum et al., 2003), and ORYZA (Bouman, 2001) used climate, soil, and ecophysiological crop parameters to simulate yields. Simulation of crop yields is

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affected by human decisions, forcings such as temperature (i.e., growing degree days, GDD) and precipitation rate/pattern, planting date, irrigation and fertilizer amount and timing, crop cultivars, and atmospheric CO_2 (Baigorria et al., 2008). Among all these, PD has a significant impact on the crop model output (Urban et al., 2018), owing to its effect on growth duration and critical phenological phases. For rainfed paddy in particular, the PD and the length of the wet and dry season as well as the timing of precipitation can influence growth, physiology, and estimated yield (Tsimba et al., 2013). Several studies have shown that late sowing reduces yields whereas timely sowing increases crop production (Irwin and Hubbs, 2019; Kucharik, 2008). In almost all crop growth models, soil conditions, climatology, and water availability are used as inputs to calculate PD. Moreover, crop calendars developed from long-term PD observations provide fixed PD that do not account for the current field conditions, weather events, and socioeconomic factors that may lead to divergence of PD within the study region (Zhang et al., 2021). Hence, for a particular location or region of interest, getting information on PD data purely by any one indicator is difficult and may lead to erroneous results and adversely impact the crop modeling performance. Furthermore, the choice of cultivars with long or short durations also plays a major role in PD selection (Mandal et al., 2018). Thus, to have an optimal estimation of yield through crop models, it is necessary to have realistic near real-time information on PD.

Recent advances in remote sensing have provided scientists and resource managers with the opportunity to estimate PD using near-real-time satellite observations that captures the farmer's decisions/interventions into crop models to forecast yields. In remote sensing techniques, microwave, optical, and thermal sensors have provided useful tools for estimating crop PD, detecting crop types, and monitoring phenology (Boschetti et al., 2017). Satellite-based optical sensors provide muti-spatiotemporal reflectance data that can be used to determine time-series of vegetation indices in cropland areas. However, optical/thermal imagery is susceptible to clouds, aerosols, and saturation in areas with high biomass. It is also notable that in the rainfed regions and during the paddy planting period, generally overcast conditions are observed. Thus, since the 1990 s, scientists have been using and exploring microwave SAR observations at finer resolution (~10 m) with day-night and all-weather capabilities for crop monitoring.

Studies have been conducted recently to evaluate the potential of SAR observations in different frequency bands for rice identification and phenology detection, including the L-band (Wang et al., 2005), C-band (Canisius et al., 2018a), and X-band (Küçük et al., 2016). The results showed that C-band and higher frequencies are more capable of crop attributes detection since lower frequencies penetrate through crops/vegetation and also interact with the underlying layers (soil and water) (Friesen et al., 2012).

SAR backscatter carries information on structural and dielectric properties of the vegetation canopy that may be unique to each crop class, providing valuable information for phenology tracking and crop discrimination (Steele-Dunne et al., 2017). Additionally, the interaction between the SAR backscatter and the rice canopy depends on the electromagnetic polarization. A co-polarized signal like VV is particularly sensitive to the vertical orientation of the leaves (McNairn et al., 2009b), while a cross-polarization channel like VH creates a stronger correlation with the Leaf Area Index (LAI) resulting from the volume scattering within the crop canopy (McNairn and Brisco, 2004). With the launch of the Copernicus Sentinel-1 satellites in 2014 (Sentinel 1A) and 2016 (Sentinel 1B), the multi-temporal, dual-polarized, C-band backscatter images with ~ 10-meter spatial resolution and 6–12 days revisit time paved a new approach of mapping rice fields and tracking growth/phenology. Based on Sentinel-1 polarimetric observations, the following methods has been previously used to estimate the rice PD and growth cycles: (i) Using backscatter intensities that show unique characteristics of rice crop (flooded stage, low backscatter of water) (Yang et al., 2021), (ii) Using machine learning crop classification and phenology detection techniques (Lasko et al., 2018), and (iii) Using plant growth indices as a function of co-polarized and cross-polarized backscatter intensities to track the crop growth cycle (Mandal et al. 2020).

Our study uses the Sentinel-1A backscatter time-series analysis to fairly estimate the accurate PD for rice crops within a season. We particularly focused on the assessment of SAR PD impact on crop model yield estimation, as there have been no studies that examine this topic. The specific objectives of our study are as follows: (i) To map rice planting areas using analytical time-series analysis of Sentinel-1A, (ii) To determine the observed PD through tracking of the backscatter time-series, and (iii) To evaluate the impact of the SAR-derived PD on physically-based crop model performance in estimating yield.

2. Study area, Datasets, and tools

2.1. Study area

Rainfed paddy fields in Cambodia were chosen as our study area (Fig. 1a). As a tropical country, Cambodia has a high average temperature with dry season prior to wet rainy seasons. The rainy season is from May to October, with ~ 75–80 % of the total annual precipitation is brought by south-westerly summer monsoon and the dry season occurs from November to April (Nguyen et al., 2010).

The majority of the paddy in Cambodia is rainfed and cultivated during the wet season (Wang et al., 2017). With ~ 2.9 million acres of rice cultivated and ~ 11 million tons of production, Cambodia is ranked tenth among the global rice producers in 2020 (FAO, 2020). Seven provinces reported planting over 100,000 ha: Prey-Veng, Battambang, Banteay-Meanchey, Takeo, Kampong-Thom, Siemreap, and Svay-Rieng (NIS, 2019). In the study, yield simulations were conducted for these provinces. In Cambodia, fertilizer use is much lower (~34.3 kg/ha in 2018) compared to the other neighboring Southeast Asian countries (World Data ATLAS).

2.2. Datasets

2.2.1. Sentinel-1A (SAR) data

Alaska satellite facility (ASF) provides free access to Sentinel-1A IW level 1, ground range detection (GRD) data with a 12-day repeat cycle of the C-band SAR. For this study, Sentinel-1A radiometrically terrain corrected (RTC) products generated by ASF using GAMMA software (Copernicus Sentinel data, 2022) were downloaded (from 2017 to 2020) for three different ascending tracks covering Cambodia (Fig. 1a).

2.2.2. Satellite and ancillary data used for rice yield estimation

Table-1 provides information related to precipitation, temperature, wind speed, crop calendars, crop maps, soil maps, rice yield observations, and fertilizer data used in this study. This study used a crop calendar developed by the Center for Sustainability and the Global Environment by compiling data on rice planting and harvesting dates from six different sources such as USDA-FAO, USDA-FAS, USDA-NASS, and IMD-AGRIMET (Sacks et al., 2010).

2.3. Rice crop model

We implemented the Regional Hydrologic Extremes Assessment System (RHEAS) framework (Andreadis et al., 2017) over our study domain. It is a comprehensive drought and crop yield information system that loosely couples a hydrologic model (Variable Infiltration Capacity, VIC) with a crop model (Modified Decision Support System for Agrotechnology Transfer, M−DSSAT) (Ines et al., 2013). RHEAS allows for the investigation of the effects of weather conditions, hydrology, climate, and farm management practices (e.g., PD, fertilizer rates) on crop yield. Using RHEAS, a regional based study has been previously carried out by Abhishek et al. (2021) that evaluated the potential implications of growing season drought on interannual variability of rice yields. Table 1 presents a list of the meteorological forcings, soil

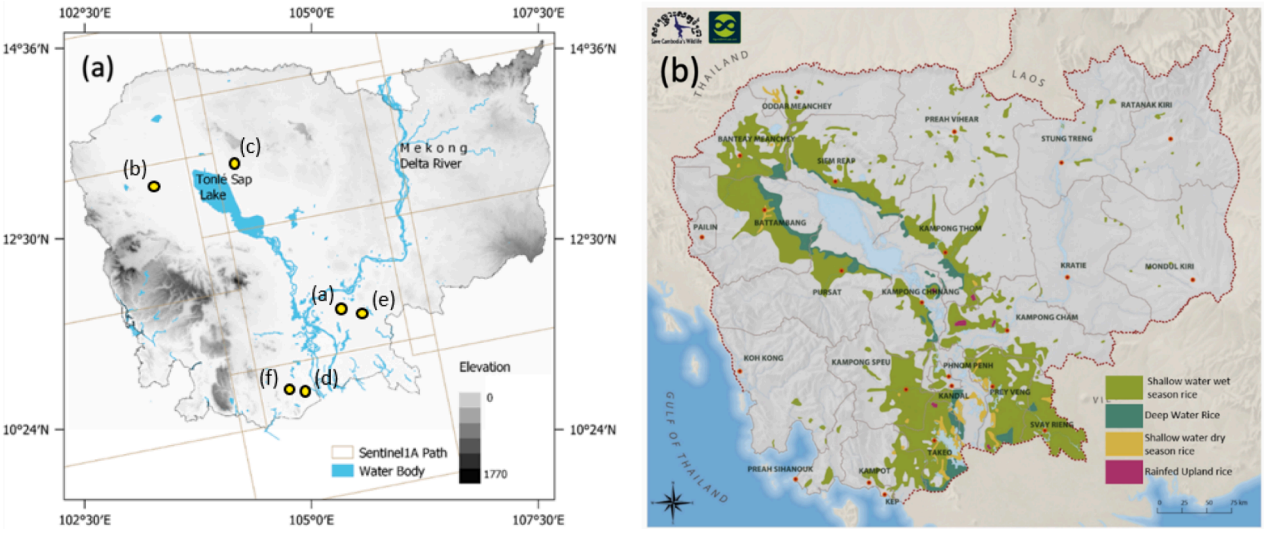


Fig. 1. a) Study area (Cambodia), the background in the map is the 30 m DEM. The rectangular tiles are the Sentinel-1A tiles showing three ascending satellite tracks covering Cambodia. The yellow points are showing the location of the selected pixels in Fig. 3, b) Region with rice crop in Cambodia for the year 2012 from Open Development Mekong platform (https://bit.ly/3tFPuCI). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 1
Satellite and ancillary data used for the study.

Parameter	Product	Spatial resolution	Temporal resolution	Temporal coverage	Reference
Precipitation	Chirps	5 km	daily	1981-present	(Daly et al., 2008)
Temperature	NCEP	1.875 deg	daily	1948-present	(Kanamitsu et al., 2002)
Wind speed	NCEP	1.875 deg	daily	1948-present	(Kanamitsu et al., 2002)
Soil type	SoilGrid250m	250 *m*	static	2017	Open Land Map
Crop Map & Calendar	SAGE	18.5 km	static	One for all years	(Sacks et al., 2010)
SAR Backscatter	Sentinel-1A	30 *m*	12 days	2016-present	(Copernicus Sentinel data, 2022)
Fertilizer	–	One pattern	static	yearly	World Data ATLAS
Yield Observation	–	Province	yearly	2010–2020	General Directorate of Agriculture, 2020

properties, and land cover data information incorporated into the VIC hydrologic model. The hydrological outputs from the hydrologic model (specifically, rainfall, air temperature, and net solar radiation) are channelized into the crop model to simulate the crop yield. The M−DSSAT runs in an ensemble mode to capture the variability in the agricultural system due to cultivars, soil types, fertilizer and irrigation applications, and PD. In addition, RHEAS has the capability to assimilate the surface soil moisture and LAI data from different remote sensing products. Here, we calculated the range of interannual variability of rice yields within each year among different provinces using the information from SAR-derived PD. We refer the reader to Abhishek et al. (2021) for a comprehensive and general presentation of the RHEAS model.

3. Methodology

The methodology used in this study comprises four main steps: (1) Pre-processing of Sentinel-1A observations, (2) Time-series analysis of VH and VV Sentinel-1A backscatter for rice mapping, (3) Developing a PD retrieval algorithm using VH Sentinel-1A backscatter time-series, and (4) Yield estimation using SAR-derived PD in the RHEAS framework.

3.1. Pre-processing of Sentinel-1A images

RTC images processed through the Gamma software in ASF were downloaded at a 30-meter resolution. We used the Sentinel-1A ascending mode for mapping rice and detecting PD in this study because of the negative impact of morning dew on SAR observations in tropical areas. Ten Sentinel-1A image tiles in the ascending mode were mosaicked covering wall-to-wall over Cambodia. The time difference between the three ascending tracks (Fig. 1) is 2 and 5 days. The assigned date to the mosaicked landscape is based on the average overpass dates of the three tracks. To reduce speckle noise in the VH and VV time-series, the SAR images were aggregated to 100 m using bilinear interpolation (a weighted average of the four nearest neighboring pixels). Using Equation (1), the linear VH and VV time-series were converted to decibel (dB) scale. From here on, the Sentinel-1A backscatters are denoted as σ^0.

$$\sigma^0_{dB} = 10 \times log_{10}(\sigma^0) \quad (1)$$

Where σ^0 is the backscatter at linear scale and σ^0_{dB} is in the logarithmic scale (dB). The incidence angle range of Sentinel-1A over Cambodia is 37°to46°. Rice paddy fields in Cambodia are primarily located in flat terrain (Fig. 1), which minimizes the incidence angle artifacts due to terrain (Lasko et al., 2018). Moreover, in the mosaicked landscape the incidence angle and backscatter data were taken from only one granule from the overlap region without any averaging. The detailed procedure for preprocessing the Sentinel-1A data is shown in Fig. 2.

3.2. Algorithm for rice mapping

Cambodia includes a variety of landcover types, primarily dominated by croplands, particularly for rice agriculture, urban areas, forest, and grasslands. In this study, rice mapping was conducted using $\sigma_{VH}{}^0$ and $\sigma_{VV}{}^0$ [dB] backscatter time-series. An analytical time-series analysis algorithm was developed to differentiate between rice and the other

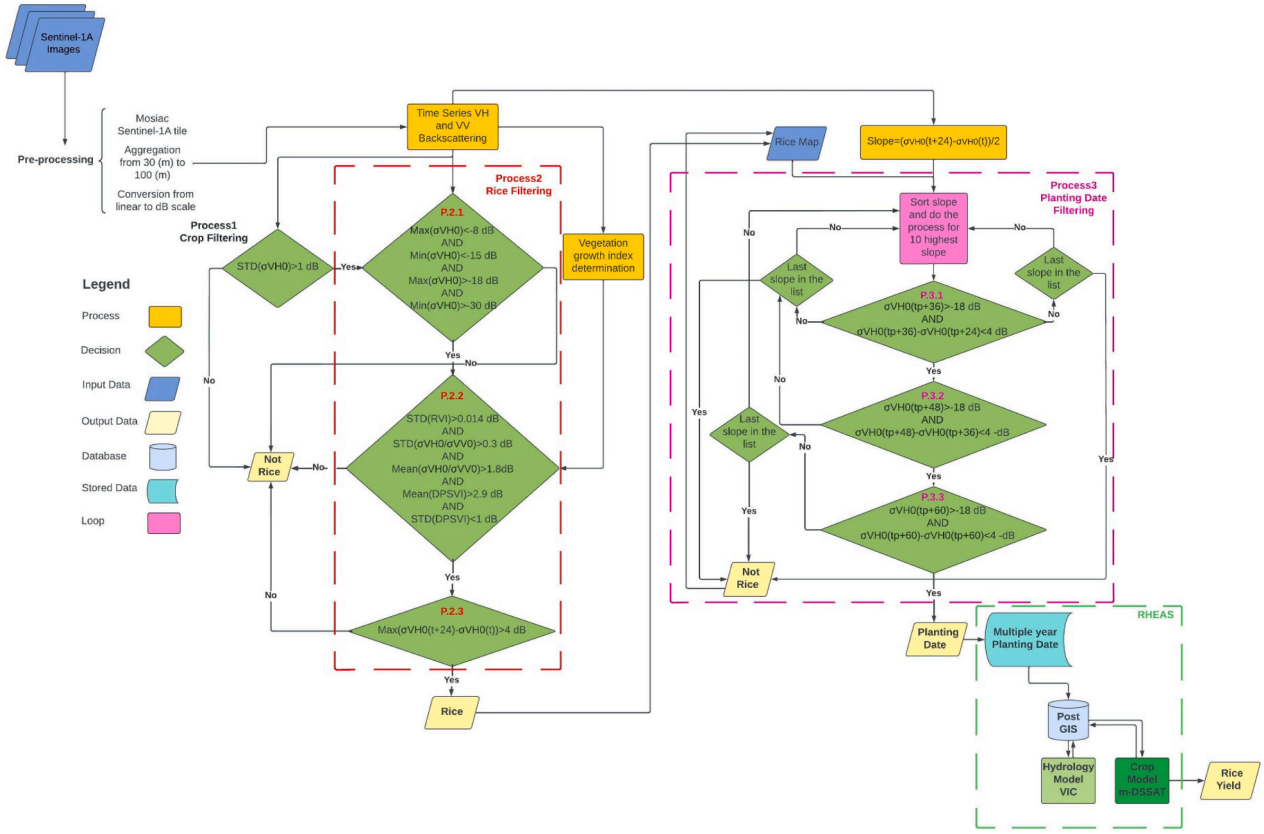


Fig. 2. Methodology for identifying rice acreage, retrieving PD, and estimating rice yield. Section 3.2 provides details on each parameter, *t* is the observed dates in $\sigma_{VH}{}^0$ time-series within the wet season, *tp* is the PD candidate, and the start time of the sorted slopes in Process-3.

crops/landcovers. In which, a set of statistical parameters were defined for cross-polarized $\sigma_{VH}{}^0$ observations, and vegetation growth indices were calculated with both $\sigma_{VH}{}^0$ and $\sigma_{VV}{}^0$ observations.

In this study, the algorithm uses cross-polarization channel ($\sigma_{VH}{}^0$) observations as the main SAR observations for both mapping rice and detecting PD. This is due to the fact that $\sigma_{VV}{}^0$ observations are influenced by standing water and largely attenuated by the rice crop, while the $\sigma_{VH}{}^0$ data are less affected and more representative of rice growth and plant canopy structure (Son et al., 2021). Additionally, Inoue et al. (2014a) found that there is a stronger correlation between $\sigma_{VH}{}^0$ and LAI resulting from the volume scattering within the crop canopy.

In a preliminary step, the rice mapping algorithm distinguishes between crop and non-crop pixels by considering a threshold for the average and standard deviation (STD) of the $\sigma_{VH}{}^0$ time-series (Fig. 2, process 1). For further differentiation between rice and other crops, four thresholds were defined for the minimum and maximum backscatter values (Fig. 2, Process 2). To choose these thresholds, a sample analysis of known paddy fields taken from landcover information was conducted to identify a temporal signature of σ^0 values over the rice-planted areas (Fig. 1b). There are three vegetation indices that were used to enhance rice detection precision: (i) co-pol ratio ($\frac{\sigma_{VH}{}^0}{\sigma_{VV}{}^0}$) (Veloso et al., 2017), (ii) Dual-pol Radar Vegetation Index (RVI) $\frac{4\sigma_{VH}{}^0}{(\sigma_{VV}{}^0+4\sigma_{VH}{}^0)}$ Nasirzadehdizaji et al., 2019, and (iii) Dual Polarization SAR Vegetation Index (DPSVI) $\frac{(\sigma_{VV}{}^0+\sigma_{VH}{}^0)}{\sigma_{VV}{}^0}$ (Mandal et al., 2020). We defined thresholds for the standard deviation and average of the above indices to distinguish the paddy pixels from the non-rice pixels using countrywide sampling. The detailed rice mapping procedure is shown in Fig. 2, Process-1 and Process-2 are elaborated below in the following steps:

Process-1: ***Differentiating between crop and non-crop***, the pixels with STD ($\sigma_{VH}{}^0$) less than 1 is classified as non-crop pixels. These pixels are mostly forest and urban areas.

Process-2: ***Differentiating between rice and the other crops***

Process-2.1: Defining a backscatter range for $\sigma_{VH}{}^0$ variation by averaging the behavior of ~ 100–200 backscatter time-series over the verified paddy fields from landcovers data (Fig. 1b). The $\sigma_{VH}{}^0$ mainly varies from −12 to −24 dB over paddy fields depending on the pixel's location in the near and far range in swath width. Due to flooding in some parts of Cambodia, the minimum $\sigma_{VH}{}^0$ may decrease to −30 dB due to the impact of specular reflection from the flooded area.

Process-2.2: Defining a threshold for STD($\sigma_{VH}{}^0/\sigma_{VV}{}^0$), Mean ($\sigma_{VH}{}^0/\sigma_{VV}{}^0$), STD(DPSVI), Mean(DPSVI), and STD(RVI) using the countrywide sampling to increase the accuracy of rice differentiation from the other crops.

Process-2.3: Eliminating the pixels in which the highest rise in $\sigma_{VH}{}^0$ time-series is less than 4 dB in 24 days (two consecutive Sentinel-1A observations). 4 dB was determined by averaging the behavior of the $\sigma_{VH}{}^0$ time-series over the verified rice fields from landcovers (Fig. 1b).

3.3. Planting date detection

By considering the unique phenology of the paddy σ^0 signals as well as a general time period for the growing seasons (e.g., May to August is the wet season in Cambodia), we found the local PD coinciding with the local minimum value of the σ^0 time-series, indicating an inundated surface. Both sowing and transplanting methods for rice planting, result in a double bounce at the beginning of the growth cycle that causes an increase in the σ^0 time-series. In the case of transplantation, we would experience the double bounce immediately after transplantation. For sowing, after 15–20 days, the plant height is 15–20 cm with 2–3 leaves, and this is the time when the soil is flooded with water and a sharp rise is observed in the σ^0 time-series. Therefore, as the beginning of the rise in

σ^0 time-series is 12 days before the double bounce occurrence, it can be a good representative of the sowing time which is the most commonly used rice planting method (Kamoshita et al., 2016). It is notable that the water droplet on the rice leaves can also increase the radar backscatter intensity up to 2 dB in co– and cross-polarized backscatter timeseries (Khabbazan et al., 2022), however this increase is negligible compared to the sharp rise due to the planting double bounce effect.

Most of the rice fields in Cambodia are rainfed and no data is available on rice PD, thus a larger time window from March to August was considered in order to determine PD in various provinces. It is easier to determine the PD over the irrigated paddy fields compared to the rainfed rice fields due to the distinct seasonal rise and fall pattern in σ^0 time-series in irrigated rice fields. Rainfed areas are usually plowed to a depth of 70–100 mm (depending on the soil condition), and they may need to be plowed again within 3–6 weeks after the initial plowing which results in an increase in soil surface roughness. Consequently, the σ^0 time-series varies frequently before planting in rainfed areas, which can be mistaken as a planting signal. The ${\sigma_{VH}}^0$ time-series shown in Fig. 3a is from the southern part of Cambodia (next to Vietnam in which irrigated paddy field is the common practice). A pattern in the time-series indicates that irrigation is widely used in this area, making it easy to choose the local minimum or highest rise slope in the σ^0 time-series as the time for sowing. Conversely, in rainfed rice fields, the pattern of the σ^0 time-series is continuously changing (Fig. 3b-f), and it does not have a consistent pattern, making the detection of PD very challenging. Therefore, other criteria being incorporated into the algorithm for determining the increase associated with planting. Accordingly, each sharp rise in the time-series for a specific period (wet season) was examined using several criteria such as the slope and ${\sigma_{VH}}^0$ variation between 36 and 60 days after sowing. Since an average overpass date

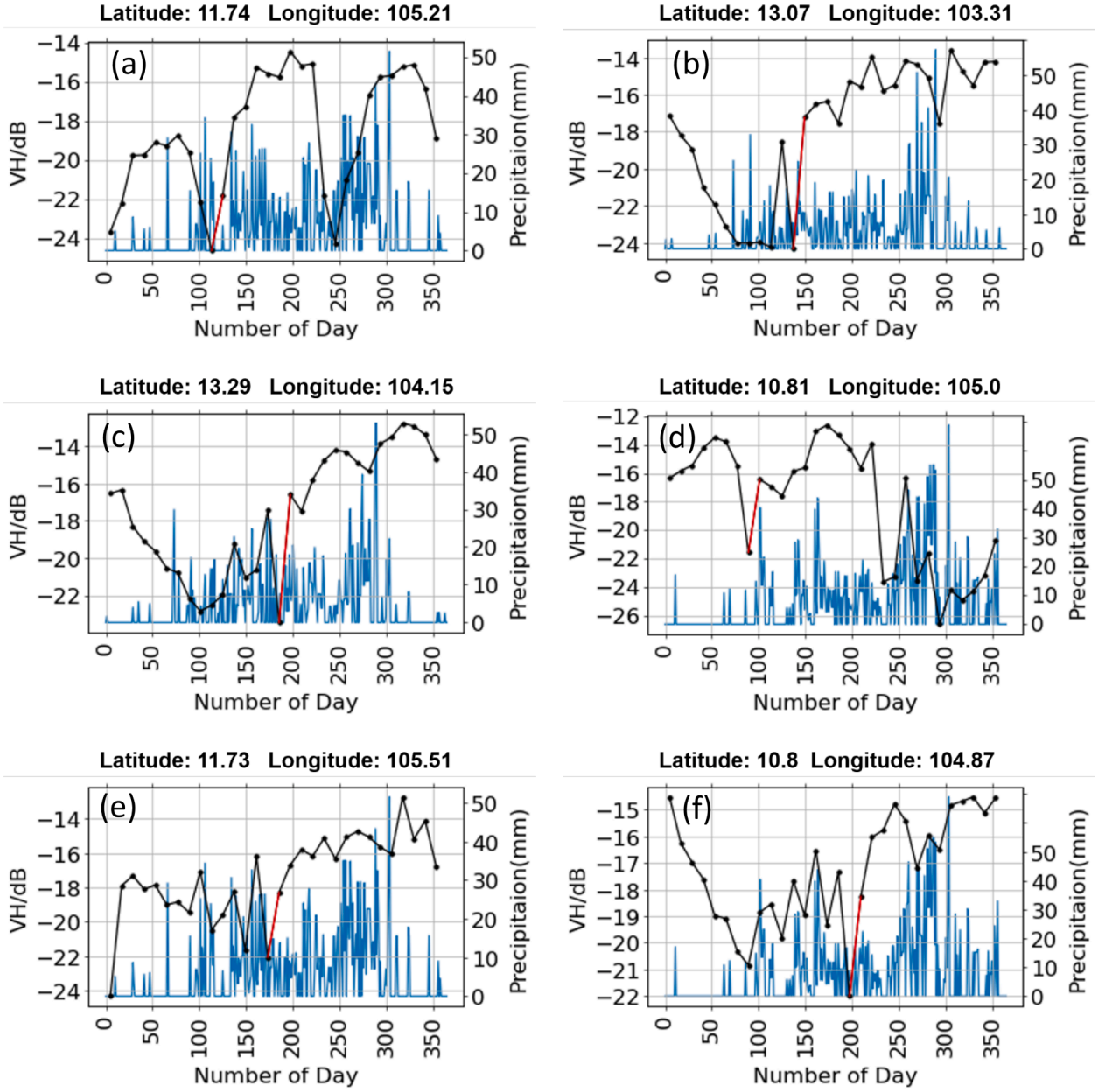


Fig. 3. Temporal backscatter profile of ${\sigma_{VH}}^0$ for rainfed rice in year 2020 for different pixels in the north and south part of Cambodia. The red line represents the sharp rise (difference between $\sigma^0_{PD+12} - \sigma^0_{PD}$) associated with the planting and the start of the rise was taken as the time of sowing. A sharp rise in the σ^0 time-series was taken as an indicator of the sowing stage. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

was considered for the mosaicked landscape of Cambodia (section 3.1), the PD for each province was corrected using the Sentinel-1A track associated with the province. The detailed PD detection procedure is shown in Fig. 2.

Process-3: ***Estimation of planting date for the rice field map determined from Process 2***

For a specific time period (March to August) all the rises in σ_{VH}^0 time-series were sorted from the highest to the lowest value. A parameter was defined to calculate the slope of each rise in σ_{VH}^0 time-series by measuring the increase in σ_{VH}^0 in 24 days. Starting from the highest slope in the list and considering the start time of the slope as the candidate PD, σ_{VH}^0 behavior was monitored 36–60 days after the candidate PD. *P*3.1 to *P*3.3 filter out PD candidates based on the criteria that σ_{VH}^0 must be less than 18 [dB] in 36–60 days after planting, and the σ_{VH}^0 time-series should not have a sharp rise higher than 4 [dB] within this period.

3.4. Yield estimation using RHEAS tool

To evaluate the efficacy of PD to estimate rice yield, the SAR-derived and crop calendar PDs were used as separate inputs to the crop model (M−DSSAT). 40 ensembles of yield scenarios were generated for each province to capture the variability due to soil, cultivars, PD, and forcings. A sequential resampling process to assign the most frequent SAR-derived PD to the ensembles in each province was applied to make sure that the most probable dates from the SAR data were used. The resampling process is analogous to rolling an *N*-sided loaded die, where N is the number of SAR-derived PD (here N = 12). The probability of rolling each side of the die is defined by weights calculated based on the fraction of the total area for each growing month (March to August) times the fraction of the month for each specific date. Hence, PDs with higher weights have higher probabilities of propagating through the ensembles. In addition to PD, fertilizer and irrigation application, and cultivar varieties also play an important role in simulating yield. To obtain a fair estimation of rice yields in each province, rice cultivar genetic coefficients must be calibrated using observation yields under no stress conditions (Boote, 1999). To calibrate the genetic coefficients, the two most common cultivars in Cambodia reported by FAO (Phka Rumduol and Sen-Pidao) were used. Phka Rumduol rice is a premium aromatic jasmine rice cultivated during the wet season with a long duration, medium maturity, and taller photosensitive rice. Sen-Pidao is a soft cooking rice with a non-glutinous aromatic flavor, short duration, and early maturity which makes it similar to jasmine rice (Nadar et al., 2020). This study used the Shuffled Complex Evolution Algorithm to calibrate cultivar coefficients. This algorithm mainly consists of multiple complex shuffling and competitive evolution based on the simplex search method (Guo et al., 2013). The simulated yield and observed yield for 11 years (2010–2020) were used to optimize the cultivar coefficients.

4. Results

4.1. Rice mapping

Our study found that vegetation indices such as RVI, DPSVI and $\sigma_{VH}^0/\sigma_{VV}^0$ have the same efficiency on rice mapping and the results improved significantly as compared to the rice map in Fig. 1b when the indices were applied to the rice mapping algorithm (Fig. S1 in the supplementary material shows the detected rice map in 2020 before and after employing RVI, DPSVI and $\sigma_{VH}^0/\sigma_{VV}^0$ in the rice mapping algorithm). While σ_{VH}^0 signal compared to $\sigma_{VH}^0/\sigma_{VV}^0$, RVI and DPSVI showed a better capability in PD detection (Yang et al., 2021). The paddy fields as shown in Fig. 4 are mostly found in low-altitude regions along the Tonle Sap Lake and Mekong River. However, rice fields are relatively sparse in the eastern part of Cambodia, which is consistent with the land cover map of Cambodia (2012) in Fig. 1b. The presence of rice fields signature closer to the lake shores (around the lake) is caused by the similarity in σ_{VH}^0 time-series between wetland and paddy fields.

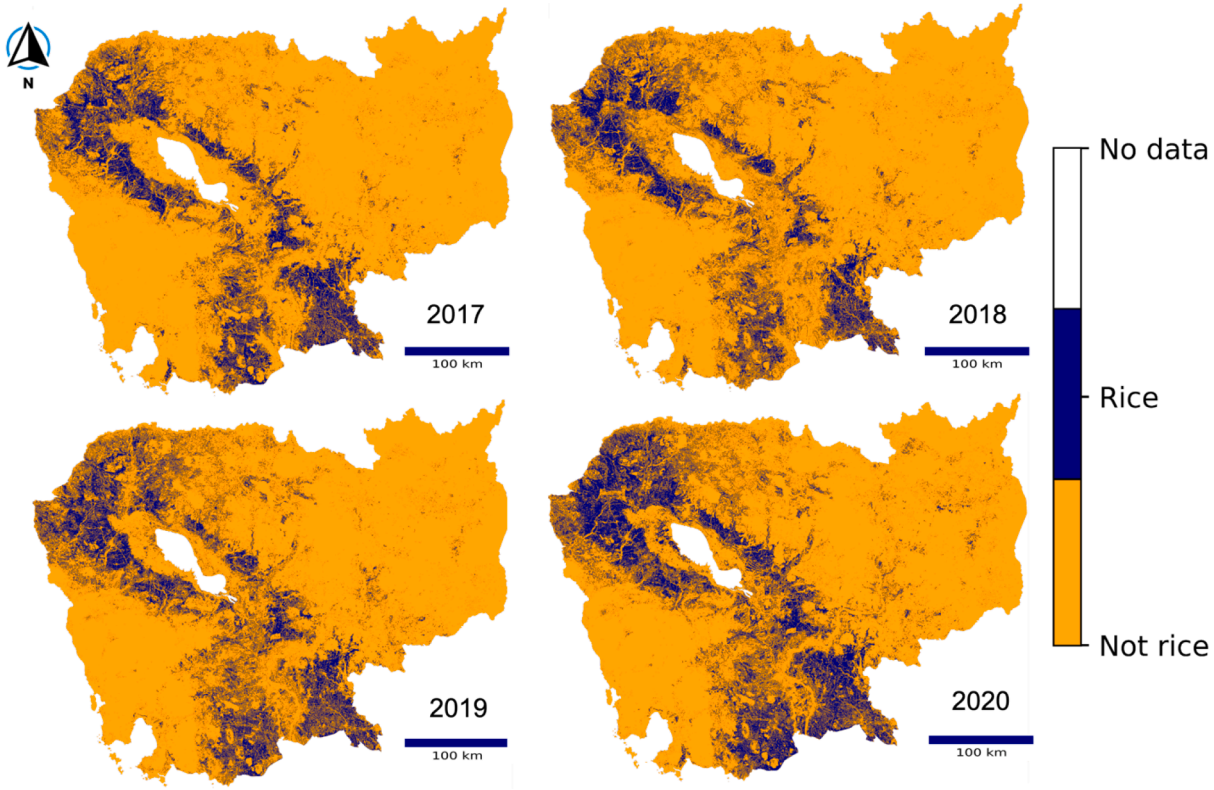


Fig. 4. Rice planted area estimated from Sentinel-1A σ_{VH}^0 observations in four years (2017–2020).

According to the landcover of Cambodia, the northwestern region close to the Tonle Sap has number of wetlands and forests. To evaluate whether the rice mapping algorithm could distinguish between crops and not crops in this study, we did not use a non-cropland mask. A comparison of rice detection map (Fig. 4) from our study to landcover map of Cambodia (Fig. 1b) shows that our rice mapping algorithm is capable of distinguishing between crops and not-crops as well as rice from other crops.

Furthermore, using the same approach we detected a significant expansion in the estimated rice planting areas in 2020 relative to the previous years (Fig. 4). The reason for detecting more paddy fields in 2020 is the false alarms due to the flooding in Cambodia between September and November 2020 caused by 13 consecutive tropical storms that resulted in unprecedented flooding and landslides in 19 provinces (Tamesis and Conan, 2020). This flooding could have resulted in spurious fluctuations in σ^0, such as many sharp peaks and valleys in σ^0 time-series same as backscatter signatures over paddy fields, wetlands, and low-lying regions.

4.2. Maps of planting dates

Fig. 5 shows the PD retrieval estimated using an analytical $\sigma_{VH}{}^0$ time-series analysis. In this study $\sigma_{VH}{}^0$ time-series were used to detect paddy PD. The dates in the legend are derived from the overpass times of Sentinel-1A, which are composited using three satellite tracks covering Cambodia. The estimated PD maps demonstrate that the PD not only changes throughout the country but also it varies within each province (Fig. 6 and Fig. 7). Seven of the 25 administrative units in Cambodia (the most cultivated areas (U.S. Department of Agriculture, 2020) were selected to examine how SAR-derived PD may vary from commune to commune and what effects dynamic PD may have on simulated yields. Fig. 6 and Fig. 7 show PD at Banteay-Meanchey in the north and Svay-Rieng in the south of Cambodia, respectively. The dates in the legend are the exact observed dates of the Sentinel-1A images before compositing three satellite tracks over Cambodia. Figs. 5, 6, and 7 collectively illustrate farmers planting rice in May thru July in 2020, while the planting season extensively varied from Mar thru Aug during 2017–2019. We hypothesize that delayed planting in 2019 (starting from April) and 2020 (starting from May) is a result of flooding in March 2019 and March and April 2020, specifically in the north of Cambodia (Banteay-Meanchey province). Inundation by flood can prevent farmers from using their ploughing machinery to prepare their fields, resulting in late planting (Fig. S2 in the supplementary material shows the area inundated in Banteay-Meanchey between March, April and May of 2017 thru 2020).

The SAR-based PD can be determined by observing the field before and after the PD in irrigated paddy fields (Yang et al., 2021), whereas in rainfed rice the σ^0 time-series needs to be monitored for 36–60 days after planting to distinguish the differences in real and spurious fluctuations (as described in the previous section) during the planting season (Fig. 3).

4.3. Impact of SAR-derived planting date on yield estimation

To test whether SAR-derived PD improves yield estimation, two simulations were conducted: 1) a data denial simulation by the M−DSSAT model using a fixed crop calendar PD (i.e., June 1 or DOY 152), and 2) a Simulation using the SAR-derived PD from Sentinel-1A observations.

The only variable changing through the 40 ensembles between the two simulations is the PD. Hence, the difference between the fixed and SAR-derived PD is the only factor contributing to the estimated yield changes through ensembles in each province. Fig. 8 and Fig. 9 show the time difference between the fixed PD from the crop calendar and the SAR-derived PD for Banteay-Meanchey and Svay-Reing provinces, respectively. Green and red areas in the figures refer to earlier and later

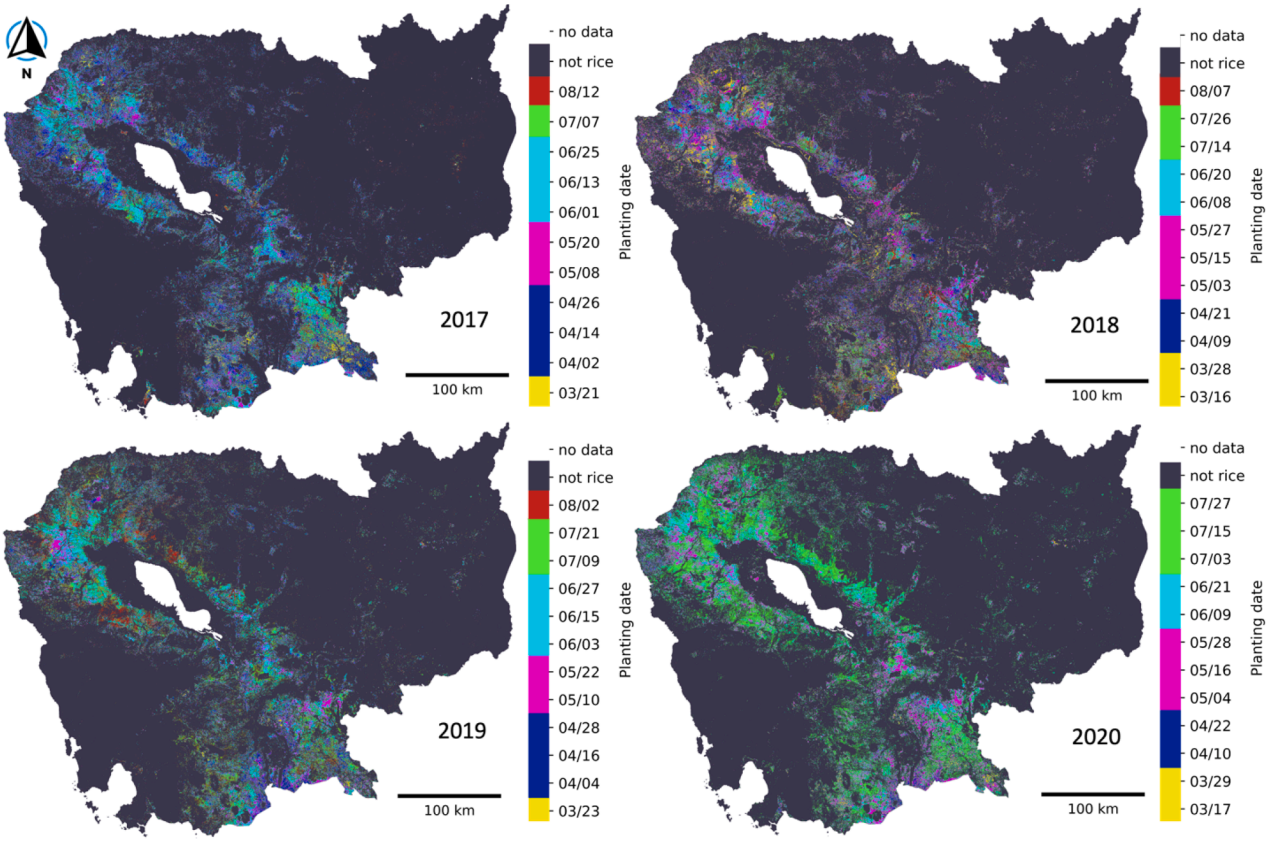


Fig. 5. The SAR-derived PD were estimated using analytical analysis of $\sigma_{VH}{}^0$ time-series for 2017–2020.

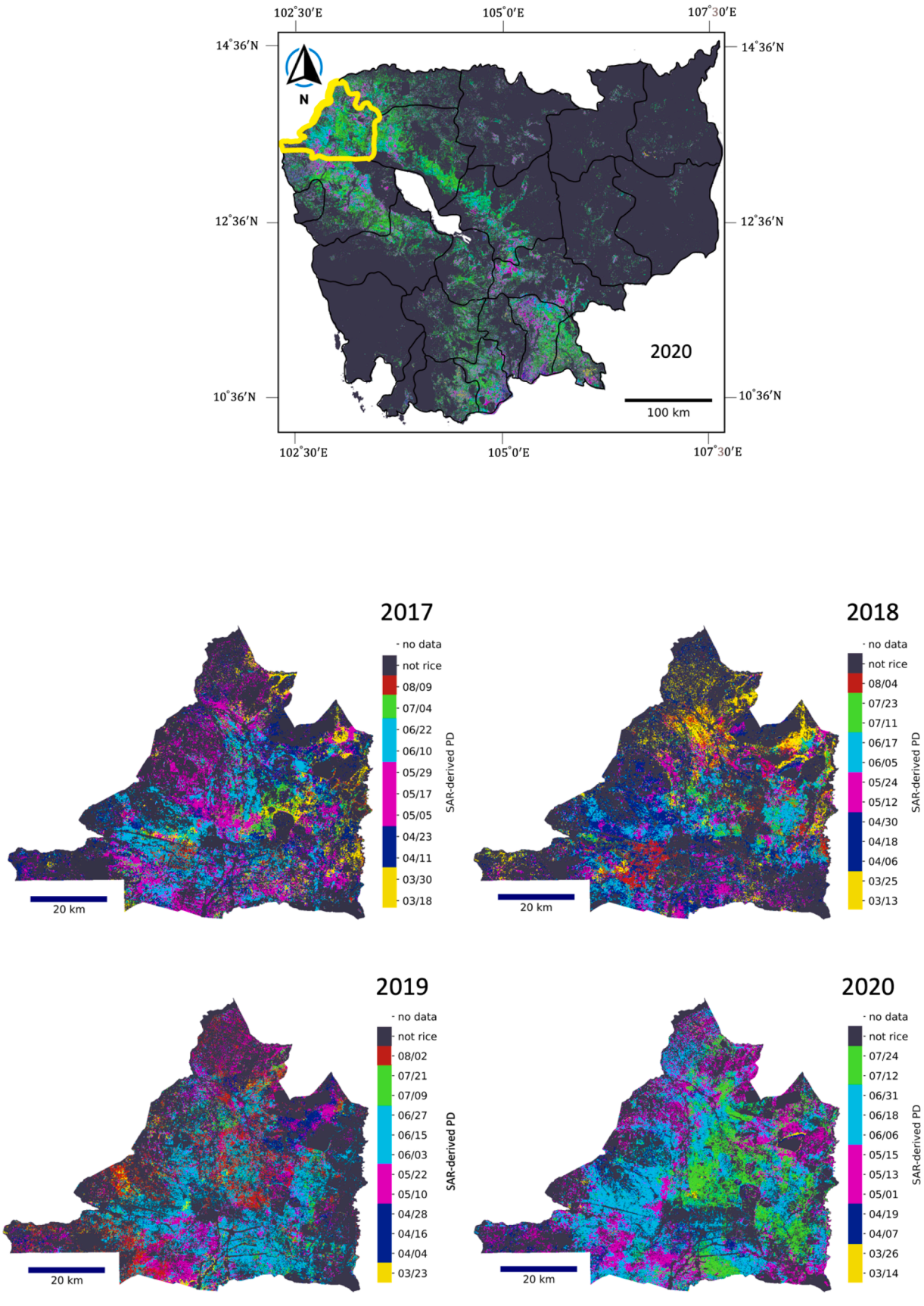
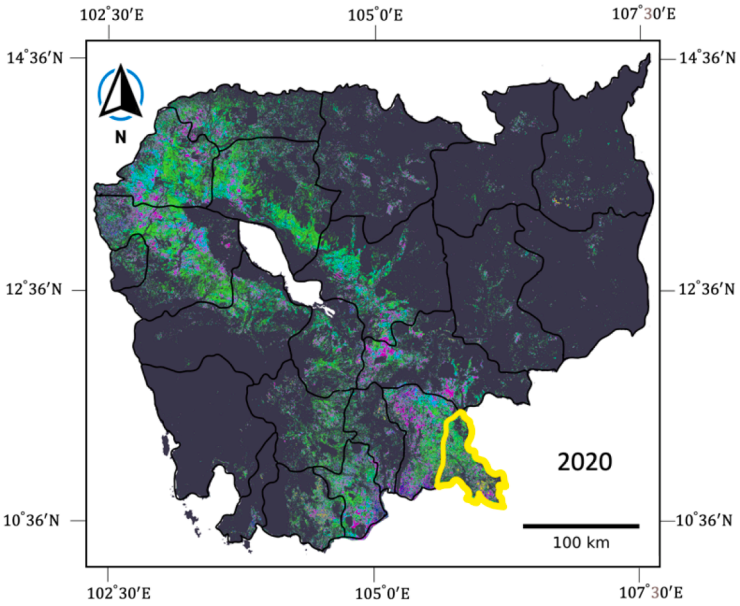


Fig. 6. Planting dates map over Banteay-Meanchey province in the north of Cambodia.



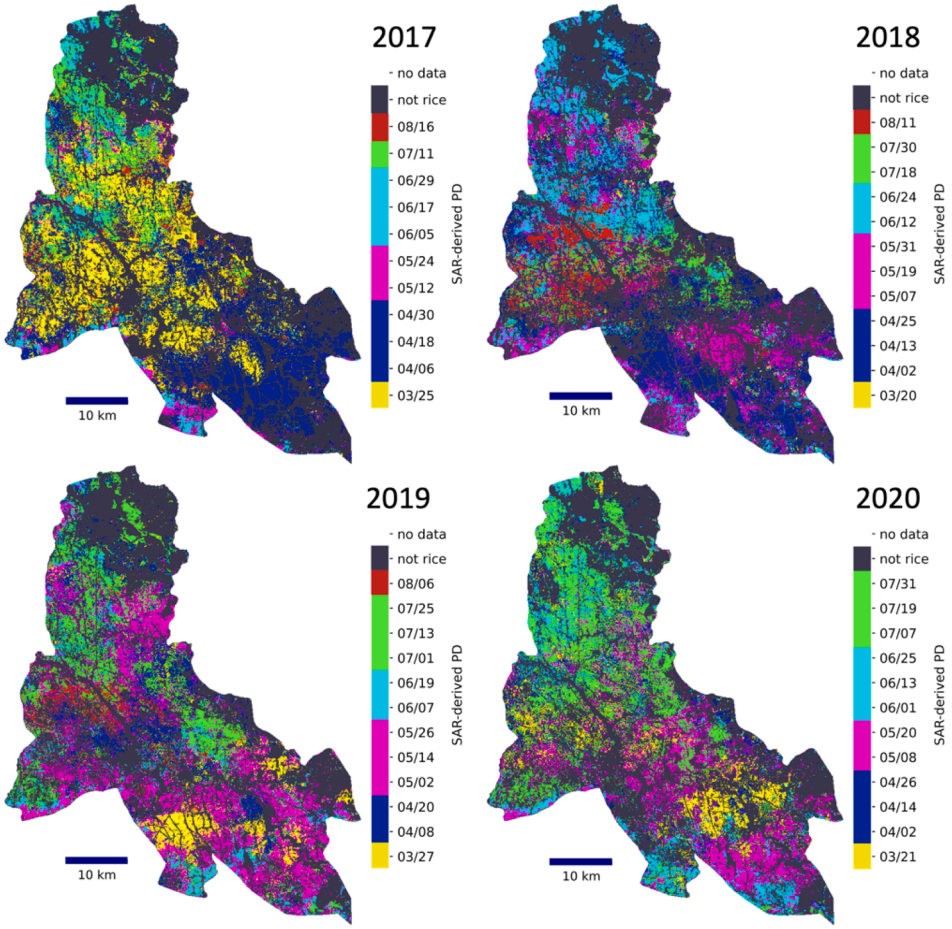


Fig. 7. Planting dates map over Svay-Rieng province in the south of Cambodia.

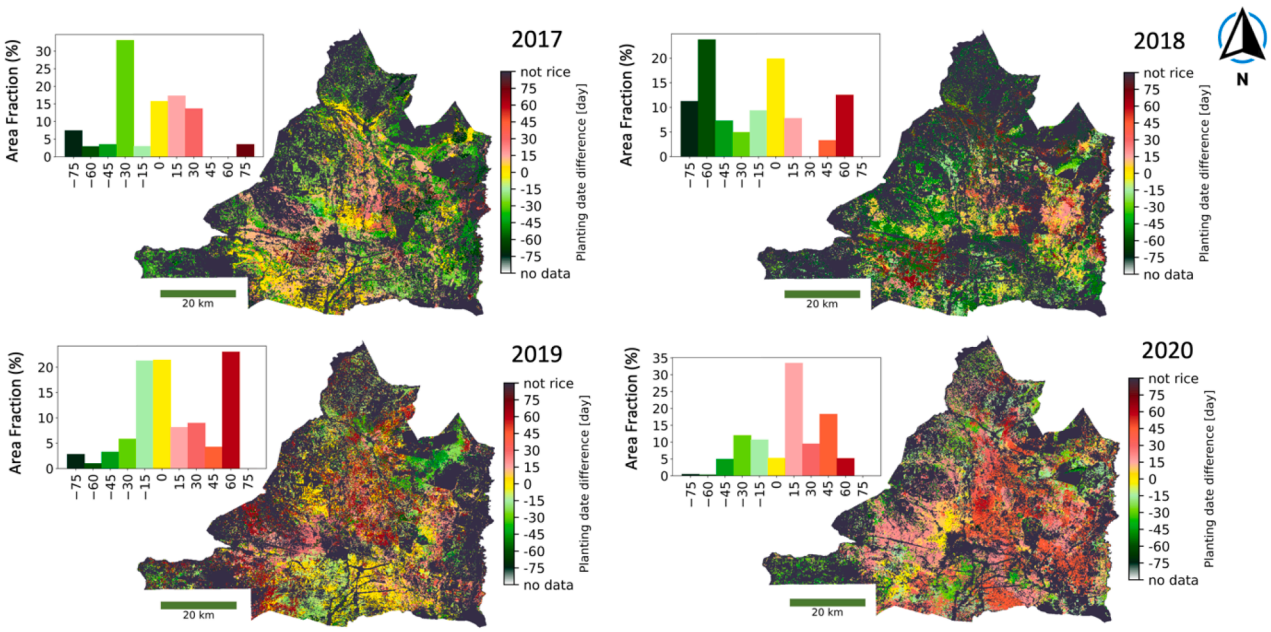


Fig. 8. The difference between the fixed PD from the crop calendar (DOY 152) and the SAR-derived PD in Banteay-Meanchey in Cambodia's north. The +/- numbers indicate planting earlier and later than the fixed date. The inset represents the fraction of total area (%) for each PD difference.

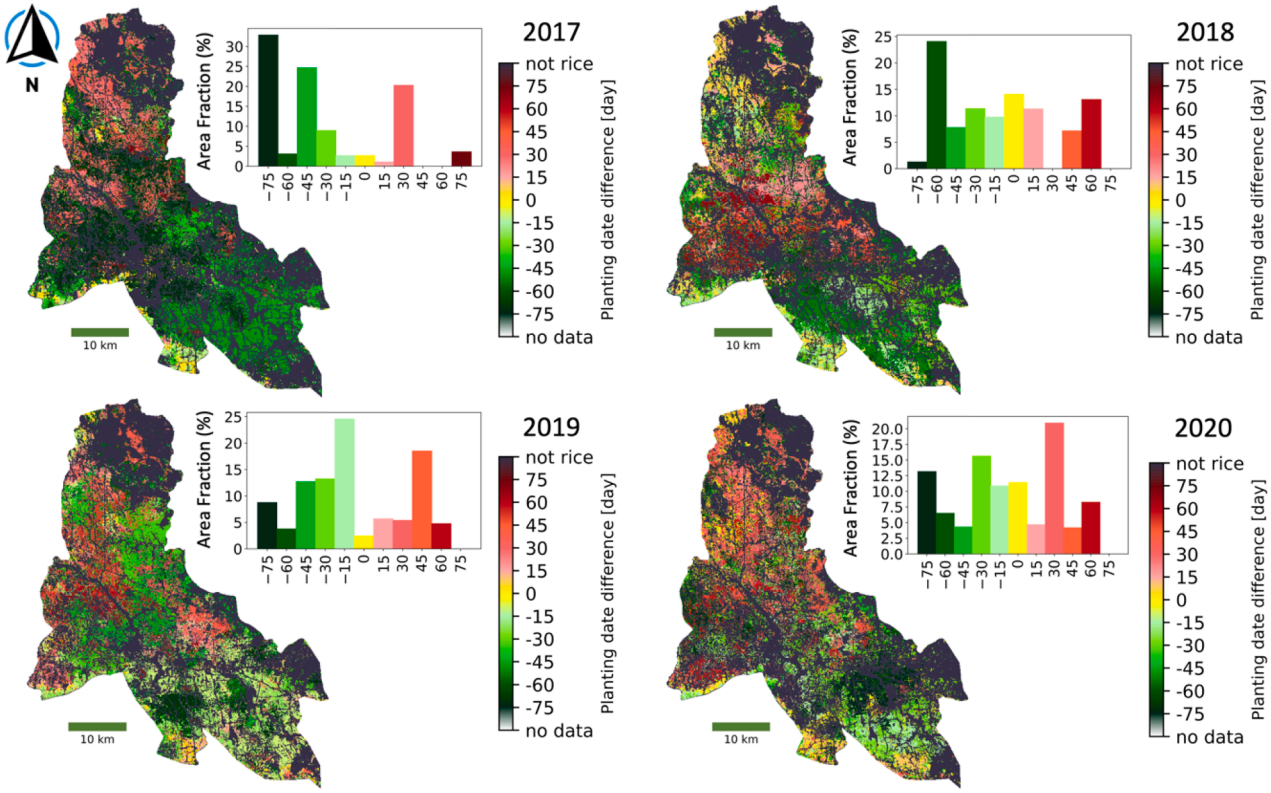


Fig. 9. The difference between the fixed PD from the crop calendar (1st of June, DOY 152) and the SAR-derived PD in Svay-Rieng in Cambodia's south.

PD, respectively, and yellow areas have an almost similar PD compared to the fixed date. According to the Fig. 8 and Fig. 9, the difference between SAR-derived PD and fixed date can be as high as 75 days (earlier or later).

Green areas in Fig. 8 and Fig. 9 indicate early planting in March and April. In the Banteay-Meanchey (as an example of a north province), planting is very rare during March and April, which is to be expected from a country known for its rainfed paddy fields. However, ~30 percent of the total area over Banteay-Meanchey is planted in March thru April in 2018, primarily due to the early onset of summer monsoon. According to our investigation, Svay-Rieng, which is a neighboring province to Vietnam uses irrigation for planting rice. This fact is also supported by the backscatter time-series pattern of this province (Fig. 3a). Additionally, the March and April planting has declined in

both provinces from 2017 to 2020. A 30 % planting in March and April between 2017 and 2018 (-75 to −60 days PD difference) was declined to 15 % between 2019 and 2020 in Svay-Rieng (Fig. 9). Changes in weather and lack of irrigation water could contribute to this decline.

Fig. 10 and Fig. 11 compare two simulated rice yields using a fixed PD and SAR-derived PD over Banteay-Meanchey and Svay-Rieng, respectively. In both simulations, all other input parameters were kept the same through the ensembles, such as weather, soil type, crop type, cultivar coefficients, and fertilizer rate. Rather, only the fixed PD (June 1 or DOY 152) from the crop calendar was changed to the dynamic SAR-derived PD to determine the effect of the PD on rice yield estimation. The results illustrate that M−DSSAT model has mostly overestimated the rice yield using fixed PD while applying the SAR-derived PD to the M−DSSAT has improved both the average and median of simulated yield significantly. In general, estimated yield decreases as the PD moves earlier in the year and increases as the PD moves forward. Wang et al. (2017) studied the effects of shifting PD forward and backward with different fertilizer application rates on yield simulation using the CERES-Rice model. In general, they concluded that shifting PD backward and forward for Sen-Pidao and Phka-Rumduol cultivars resulted in a decrease and increase in yield respectively.

According to Fig. 10, Fig. 11, and Table 1, the normalized bias of estimated yield decreased by ~ 1.5–12 % in Banteay-Meanchey and ~ 27–40 % in Svay-Rieng in the study period after using SAR-derived PD in crop model. With a higher fraction of total area for −75 to −60 days PD difference, Svay-Rieng province's yield bias is reduced more than Banteay-Meanchey's after using the SAR-derived PD in M−DSSAT model. Fig. 8 illustrates that in 2020 the positive and negative PD difference compensate for one another in Banteay-Meanchey, and the 30 % area fraction for + 15 difference days resulted in yield overestimation and a 1.5 % increase in normalized bias (Table2). In 2017, however, the area fraction for −75 to −60 days (PD difference) is higher in this province, resulting in a 12 % reduction in the yield normalized bias compared to the fixed PD. Table 2 is showing the improvement in average estimated yield bias in all the seven provinces using the SAR-derived PD in M−DSSAT model compared to using fixed PD. Additionally, the interquartile range of estimated yield has been decreased in all the provinces by using SAR-derived PD in the crop model (Fig. 10, Fig. 11, and Table 2) resulting in a significant reduction in yield estimation uncertainty in almost all the provinces of Cambodia. For example, the uncertainty in yield estimation was reduced by 38–65 % in Banteay-Meanchey and 7–44 % in Svay-Rieng.

5. Discussion

5.1. Capabilities and limitations of SAR in acreage and planting date detection

We demonstrated the utility of SAR imagery for the detection of lowland rice, particularly in tropical and subtropical regions where widespread cloud cover and complex cropping systems and land use changes (e.g., forestation) prevail, thus preventing the use of optical imagery. Rice mapping in Cambodia has become feasible due to the high spatial resolution of SAR images (<=100 m) as the paddy fields are typically less than two hectares and, in some cases, less than one hectare in this region (Pandey et al., 2010). Moreover, as a result of flooding, soil moisture and surface roughness were no longer a factor affecting backscattered SAR signals in rice fields, which has made SAR observations more effective in determining rice PD.

For accurate detection of rice fields, the number of SAR acquisitions should cover the whole season that lasts three to four months (Nelson et al., 2014). On the other hand, from the perspective of data mining, using all images for one year would maximize rice mapping accuracy.

The SAR-based PD can be determined by utilizing two images before and after the PD in irrigated paddy fields (Yang et al., 2021), whereas in rainfed rice-planted areas such as in Cambodia the backscatter time-series need to be monitored for ~ 36–60 days after planting to distinguish the actual increasing trend (caused by rice planting) and spurious fluctuations (frequent fall and rise due to preparation of the field before planting or some other reasons as illustrated in Fig. 3).

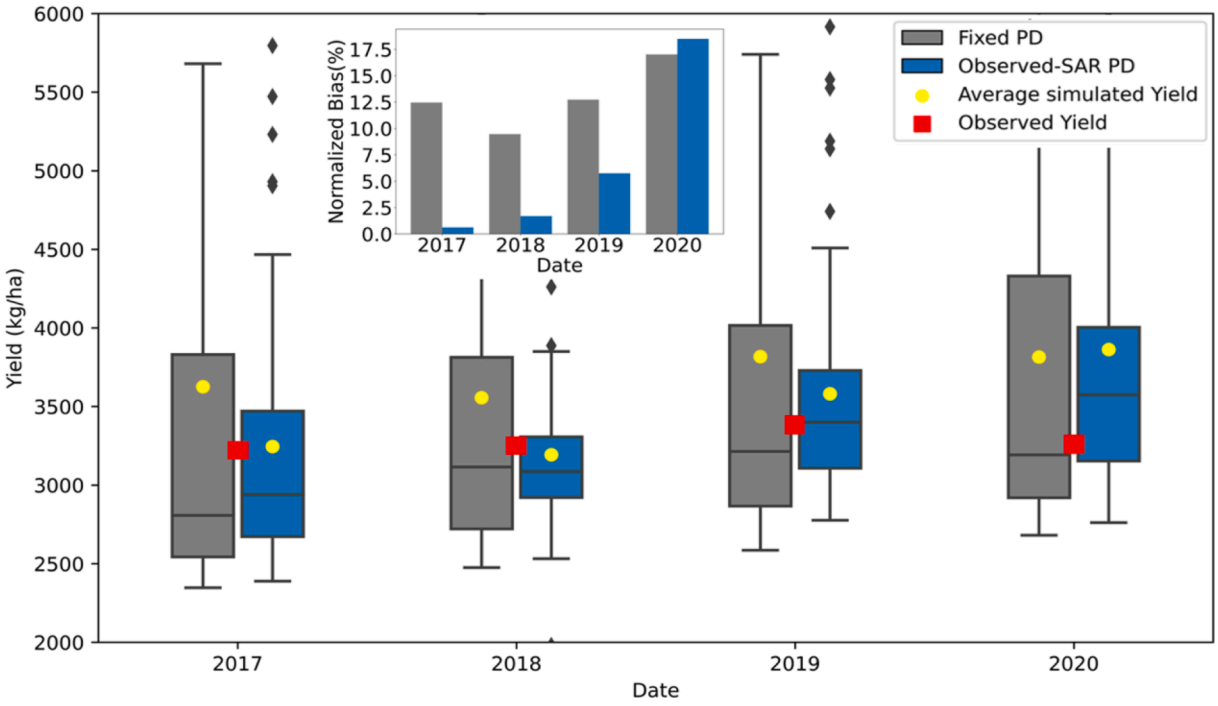


Fig. 10. Estimated rice yield in 2017–2020 in Banteay-Meanchey province. The gray and the blue boxes are showing the estimated rice yield using fixed and dynamic SAR-derived PD, respectively. The normalized bias was calculated using the average value of the simulated yield. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

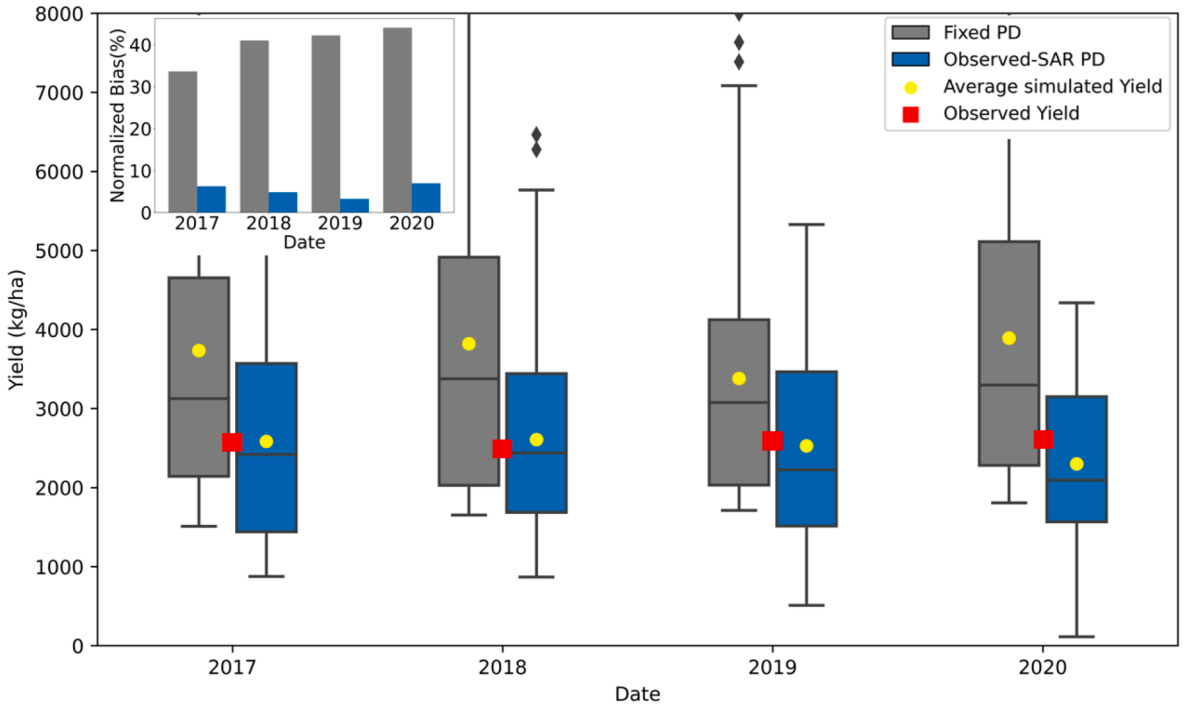


Fig. 11. Estimated rice yield in 2017–2020 in Svay-Rieng province. The gray and the blue boxes are showing the estimated rice yield using fixed planting and the dynamic SAR-derived PD, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

5.2. Impact of 12 days Sentinel revisit time on SAR-derived planting date accuracy

The yearly variations of SAR-derived PD in Fig. 5 are related to the shifting of the monsoon season (Loo et al., 2015). Due to the unpredictable nature of the monsoon onset and pattern, rice yield estimation using the SAR-derived PD can provide a more optimal estimate of yield. Moreover, the tropical and subtropical climate in much of Asia means that rice can be cultivated with diverse cropping calendars and practices over very short distances (Nguyen et al., 2012). Moreover, each province in Cambodia has some variations in its planting schedules, in addition to the annual changes. Considering the fact that farmers' decisions for planting are likely based on rules of thumb that include what their neighbors' whereabouts (Zhang et al., 2021), the PD clusters can be observed across each province (Fig. 6 and Fig. 7).

Since, the Sentinel-1A revisit time is 12 days, it is not possible to determine the exact date of the planting using σ^0 time-series. The wall-to-wall coverage of Sentinel-1A granules over Cambodia was achieved by compositing the observations over 12 days, and the composited observation map assigns an average date to compute the first and the last dates of the 12 days interval. This approach ensures that the planting date uncertainty for a given pixel at 100 m resolution can be <= +/- 6 days. The 12-day overpass interval can be improved to 6-days by including descending Sentinel-1A overpasses. However, this complicates the PD retrieval algorithm due to different incidence angles and azimuth of Sentinel-1A descending overpass, as the algorithm performance depends largely on backscatter observations that have exact incidence angle and azimuth. It is also possible to improve the revisit interval by using a combination of Sentinel-1A and Sentinel-1B that leads to SAR backscatter within 6 days of revisit time. However, in this case, also, the difference in incidence angle and azimuth would reduce the performance of our algorithm. Moreover, Sentinel-1B is unavailable since 23 December 2021 and the combined product of Sentinel-1A and 1B cannot be used in the future (European space agency, 2022).

We used an indirect approach to validate the SAR-derived planting date through comparison with the rice yield data for the provinces in Cambodia. Our study demonstrated the significant improvement in the bias and RMSE of rice yields after using the SAR-derived planting dates instead of conventional use of crop calendar in modeling rice production over the whole study area.

5.3. Rice yield estimation

Table 2 shows the reduction in both normalized bias and the interquartile range (uncertainty) of the estimated yield as the impact of using SAR-derived PD in a crop model. A fixed PD in the M−DSSAT model can cause a wide range of crop yield estimations since the PD is constant for all the ensembles, which means that actual precipitation and temperature (i.e., water availability and growing-degree-days) for the crop growth duration may not actually represent with the use of fixed PD, resulting in an under- or overestimation of yield. While a SAR-derived PD is an observation of the farmers' decision to plant and represents the near actual ground conditions. As a consequence, the use of fixed PD in the crop model increased the number of outliers and interquartile range in all provinces, ultimately increasing the uncertainty of yield estimation.

The irrigation application in the southern provinces led to the rice experiencing two or three growing seasons each year, which shifted the PD of the wet season to March and April, whereas the fixed PD is June 1st (DOY 152). Thus, the M−DSSAT model overestimated rice yield in the southern provinces (Prey-Veng, Svay-Rieng, and Takeo) using a fixed PD due to the impact of late planting on estimated yield (Wang et al., 2017). Fig. 8 shows that half of the Banteay-Meanchey was planted in July in 2020, which resulted in an increase in yield estimation using SAR-derived PD.

Another factor that is responsible for the variation in yield estimation is considering two different cultivars for 40 ensembles in each province. Depending on the cultivars used for each ensemble, PD has a different impact on yield estimation. Sen-Pidao, for example, shows a larger reduction when planting shifted earlier to March and April than Phka-Rumduol does. The variation in yields was less correlated with PD at Phka-Rumduol than it was at Sen-Pidao.

Table 2
Mean estimated rice yield in 2017–2020 using fixed and SAR-derived PD for seven provinces in Cambodia. The positive bias difference is showing an improvement in yield estimation using SAR-observed PD compared to using fixed PD in crop model and the negative bias difference is illustrating the increase in normalized bias.

Year	Province	Yield (kg/ha) Fixed PD	Yield (kg/ha) SAR derived PD	Yield (kg/ha) Observation	Normalized bias % (Fixed PD)	Normalized bias % (SAR derived PD)	Bias difference %	Interquartile range (Fixed PD)	Interquartile range (SAR derived PD)	Uncertainty Reduction %
2017	Banteay Meanchey	3626.2	3244.9	3224.5	12.5	0.6	+11.8	1287.3	798.0	+38.0
	Battambang	2945.1	3080.4	2976.6	−1.1	3.5	−2.4	1532.3	1506.0	+1.7
	Siemreap	3614.5	2639.6	2581.5	40.0	2.2	+37.8	3110.5	2050.3	+34.1
	KampongThum	3140.7	2554.9	2659.6	18.1	−3.9	+22.0	1728.0	1033.3	+40.2
	Prey Veng	4135.5	3596.5	3100.1	33.4	16.0	+17.4	3519.8	2690.0	+23.6
	Svay Rieng	3733.8	2583.7	2578.2	44.8	0.2	+44.6	2511.0	2128.8	+15.2
	Takeo	3592.8	3097.5	3029.4	18.6	2.3	+16.4	2643.8	1427.5	+46.0
2018	Banteay Meanchey	3555.8	3193.0	3248.1	9.5	−1.7	+7.8	1091.5	385.5	+64.7
	Battambang	3344.9	3050.1	3159.7	5.9	−3.5	+2.4	1695.0	1332.3	+21.4
	Siemreap	3948.7	2377.0	2604.7	51.6	−8.7	+42.9	3053.8	1869.3	+38.8
	KampongThum	2741.5	2592.3	2756.3	−0.5	−5.9	−5.4	1583.0	765.0	+51.7
	Prey Veng	4222.5	3211.2	3207.8	31.6	0.1	+31.5	3962.0	2440.5	+38.4
	Svay Rieng	3820.9	2607.1	2487.2	53.6	4.8	+48.8	2885.5	1755.0	+39.2
	Takeo	3546.9	3371.3	3071.4	15.5	9.8	+5.7	1715.8	1208.8	+29.6
2019	Banteay Meanchey	3817.5	3580.9	3386.1	12.7	5.8	+7.0	1150.0	621.5	+46.0
	Battambang	3314.1	3050.1	2947.8	12.4	3.5	+9.0	1458.3	1451.8	+0.4
	Siemreap	3182.9	3068.2	2367.3	34.5	29.6	+4.8	1896.5	2306.3	-21.6
	KampongThum	3024.8	2641.9	2695.9	12.2	-2.0	+10.2	1373.0	864.0	+37.1
	Prey Veng	3691.0	3155.7	3250.0	13.6	-2.9	+10.7	2351.0	2253.0	+4.2
	Svay Rieng	3380.0	2527.7	2567.7	31.6	-1.6	+30.1	2091.0	1951.3	+6.7
	Takeo	3472.7	3182.2	3082.0	12.7	3.3	+9.4	1473.3	1644.8	-14.4
2020	Banteay Meanchey	3815.0	3862.8	3259.9	17.0	18.5	-1.5	1411.0	850.3	+39.7
	Battambang	3768.9	3180.7	3038.1	24.1	4.7	+19.4	2051.0	1413.3	+31.1
	Siemreap	3555.3	3056.4	2770.5	28.3	10.3	+18.0	2788.3	2177.8	+21.9
	KampongThum	2923.6	2844.9	2837.6	3.0	0.3	+2.8	1309.3	1012.3	+22.7
	Prey Veng	4272.2	3079.4	3343.5	27.8	-7.9	+19.9	3669.3	2178.8	+40.6
	Svay Rieng	3891.6	2301.0	2585.0	50.5	-11.0	+39.6	2829.8	1583.5	+44.0
	Takeo	3863.2	3239.0	3093.3	24.9	4.7	+20.2	2496.5	1532.8	+38.6

6. Conclusion

In this study, an analytical time-series analysis algorithm was developed based on Sentinel-1A backscatter time-series to detect rice fields and the PD in a paddy-dominated region. The study aimed to investigate the impact of using SAR-derived PD in a physically-based crop model to estimate rice yield.

The results showed a significant departure of the SAR-derived PD from the crop calendar dates that are based on climatology (+/-75 days). According to our findings, detection of PD from SAR backscatter time-series analysis of rainfed paddy fields requires more consideration than that of irrigated rice fields. The Sentinel-1A σ_{VH}^0 time-series has been shown to improve the accuracy of PD detection in rainfed paddy areas while applying the $\sigma_{VH}^0/\sigma_{VV}^0$ ratio to the algorithm improved the accuracy of rice mapping.

The estimated yields were significantly improved by replacing the fixed PD from crop calendar with the SAR-derived PD. The average bias improved by 7–12 % and 30–48 % and the uncertainty in yield estimation was reduced by 38–65 % and 7–44 % in Banteay-Meanchey and Svay-Rieng respectively. PD retrieved in the months of March and April are from the region that is predominantly irrigated, particularly in the southeastern provinces that border Vietnam or it can be due to the early onset of the summer monsoon. As the mean yield in the rainfed lowland ecosystem is less than half of the irrigated rice yield, distinguishing between these two can help us to have an accurate estimation of rice production in the future. Using such an approach for future rice yield can help improve the overall accuracy of the total production estimations.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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